LEVELHULL · CALCULATION NOTE

LevelHull sizing calculations

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DOCUMENT CONTROL

Document ID	LVH-CAL-001	Version	0.4
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/04-calcs/01-sizing.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: four bailers (D-O3): B4 2.9 min, brim-full 11 min, kit USD 694, two canoes USD 1,384; S32 and PR-002 for the free-flooding condition (D-O2)
0.3	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: results unchanged; requirements now LVH-PRS-001 (R1 to R13 are PR-001 to PR-013); section 10 adds the STEM check results (brim-full case, pessimistic inputs, partly bailed stability, hull holes) and evidence for PR-014 to PR-017; BOM read in the kit 1.9 item format
0.2	2026-10-03	Amish Chadha	Modules widened from 200 to 220 mm (LVH-DDR-003, Amish's decision 10A); every result re-run; F20 and K3 added
0.1	2026-10-03	Amish Chadha	First issue at TRL 3, on the constructable design of LVH-DDR-002

LevelHull sizing calculations

On paper, eight foam modules of 0.42 m3 in all, fitted high under the gunwales of the 8 m reference canoe, keep the swamped canoe afloat with its motor, gear, catch and four crew holding on, with 92 mm of freeboard at the lowest point of the gunwale, level within half a degree and with 2.45 times the needed lift in reserve at 50 mm freeboard. The same foam laid low on the floor would only just reach the water surface, would heel past the 10 degree limit with the crew on one side, and would lose all roll stiffness with 5 kg more load: that comparison is why the modules sit under the gunwale. The modules were widened from 200 to 220 mm on Amish's decision of 2026-10-03 (LVH-DDR-003) to give more margin in the two weakest cases: the low-side freeboard with all four crew on one side is now 44 mm (was 31 mm) and the damaged case with the largest module lost is now 25 mm (was 9 mm). The cost is USD 384 over the value-engineering target for two canoes, with the four bailers of LVH-DEC-001 D-O3 (decided by Amish on 2026-10-25).

Every figure comes from `docs/04-calcs/sizing.py`, which imports the parametric model (`cad/src/model.py`), so the sizes here are those of the STEP files, the drawings and the build plan. Tags in square brackets match the script output and `docs/04-calcs/results.csv`. The script is also the open sizing calculator for other hulls (section 8). These are screening estimates for a paper proof of concept; they do not replace the swamp test, which is TRL 4 work.

> **Safety:** LevelHull is flotation for a boat, not lifesaving equipment, and is never a substitute for life jackets. These numbers are for a design review. No fitted canoe is used for fishing before its fixings are pull-tested and it has passed a swamp test in sheltered, shallow water with a safety boat and every person in a life jacket.

1. Method and assumptions

The swamped canoe is treated as a set of solid bodies floating in the lake. The water inside the hull is part of the lake and carries nothing. Each body gives lift equal to the water its submerged part displaces and weighs its own mass; the canoe floats where total lift equals total weight. Small heel and trim follow from the second moments of the waterplane (the strips where the modules and planks cut the water surface) and the heights of the lift and the weights: roll stiffness = water density x waterplane moment + lift x height of the centre of lift - weight x height of the centre of weight.

Table 1. Assumptions.

#	Assumption	Value	Basis
A1	Water density	1,000 kg/m3 (lakes); 1,025 kg/m3 checked	Design case is fresh water
A2	Wet hull timber density	1,000 kg/m3: no net lift from the hull	Conservative (LVH-DDR-001, D3); 750 kg/m3 reported as a margin
A3	Crew holding on	20 % of 75 kg each (15 kg), at the gunwale	Head and shoulders out of the water; four crew
A4	Outboard motor	36 kg at full weight, 0.25 m aft of the transom	15 hp two-stroke; mostly above water when swamped
A5	Gear and catch	12 kg steel anchor, 40 kg nylon nets, 150 kg fish, full fuel tank	Net in water 10.5, 4.9, 7.1 and 0 kg (a full tank floats)
A6	Foam	Closed-cell polyethylene, 30 kg/m3, low water uptake	A cited data sheet gives 33 kg/m3 and under 2 % by volume after 28 days (LVH-STM-001 S8); supplier data to confirm
A7	Kit hardware in water	Covers 9.3 kg PVC, steel 2.8 kg, straps 3.4 kg; battens neutral	Densities 1,300, 7,850 and 1,380 kg/m3
A8	Bailing	100 L per minute per person with a 10 L bucket, four bailers (LVH-DEC-001 D-O3; was two), calm water	Estimate
A9	Fixing strengths	Webbing 10 kN, D-ring pair 4 kN, M8 coach screw 1.5 kN sideways in hardwood	To confirm with real parts

2. Geometry of the kit

Table 2. Kit geometry (reference canoe).

Tag	Quantity	Value
[G1]	Side flare of the canoe	31.6 deg
[G2], [G3]	Modules	8; 4.78 m a side (880, 1,400, 1,400 and 1,100 mm)
[G4]	Module section	220 mm wide (level) x 200 mm high, a parallelogram matching the flare (was 200 x 200 mm; LVH-DDR-003)
[G5], [G6]	Module volume; foam inside the covers	0.421 m3; 0.402 m3 (was 0.382 and 0.365 m3)
[G7]	Module bottom and top above the outside of the bottom	420 and 620 mm (top 30 mm below the gunwale)
[G8], [G9]	Hull timber; dry hull mass at 750 kg/m3	0.586 m3; 439 kg
[K1]	Kit mass	68.9 kg (battens and chocks 39.4 kg, foam 12.1 kg, covers 9.3 kg); was 67.4 kg
[K2], [K3]	Cover tarpaulin; strap webbing	14.3 m2; 19.5 m

3. Swamped, design load

Table 3. Swamped canoe at the design load, fresh water.

Tag	Result	Value
[L1]	Design net load	118 kg
[F4]	Timber above the water, counted as load	109 kg
[F1]	Freeboard amidships, level	113 mm
[F3]	Trim	0.39 deg stern down
[F2]	Lowest freeboard on the gunwale (transom)	92 mm
[F5], [F6]	Metacentric height; roll stiffness	3.1 m; 761 kg m per radian
[F7], [F8]	Net load carried at 50 mm freeboard; reserve	290 kg; 2.45 times the design load
[F9]	No crew holding on	128 mm lowest freeboard, 0 deg heel, 0.14 deg trim

4. Off-design cases

Table 4. Off-design cases.

Tag	Case	Result
[F10], [F11]	All four crew holding the port grab line	3.9 deg heel; low-side freeboard 44 mm (was 4.0 deg and 31 mm)
[F12]	Salt water	98 mm lowest freeboard
[F13]	Foam lift down 5 % (the R6 limit)	86 mm lowest freeboard
[F14], [F15]	Largest module (port 2) lost, and 5 % uptake	25 mm lowest freeboard; 1.5 deg heel (was 9 mm)
[F16]	Typical 750 kg/m3 timber (lift not counted in the design)	167 mm freeboard amidships

With all four crew on one grab line the low-side gunwale is still under the 50 mm of R2, so the operating rule stays: crews spread along both grab lines. R13 itself asks only for heel within 10 deg and is met with a wide margin.

5. Why the modules are high: low placement compared

The TRL 1 concept placed the modules low along the sides. With the same 0.42 m3 of foam on the floor (85 to 285 mm up), the canoe floats with the top of the foam 1 mm under the water. The foam then cuts the waterline only by that knife edge: with four crew on one side it heels past the 10 degree limit, and 5 kg more load (one wet net, one more person's hand) puts the foam fully under water, where only the thin planks give any waterplane and the canoe has no roll stiffness left. (With the earlier 0.38 m3 of foam the floor arrangement was already in that state at the design load.)

Table 5. Same foam, high or low.

Tag	Quantity	High under the gunwale	Low on the floor
[F6], [F17]	Roll stiffness, kg m per radian	761	135, only while the foam top is at the water
[F5], [F18]	Metacentric height	3.1 m	0.32 m
[F10], [F19]	Four crew on one side	3.9 deg heel	22.2 deg heel, over the 10 deg limit
[F20]	Roll stiffness with 5 kg more load	761 or more	None (foam fully under water)

The floor figures were corrected on 2026-10-24 (v0.3): the script had used the high modules' position for the floor case's waterplane, which overstated its roll stiffness (283 against 135 kg m per radian); the correction makes the case for the high modules stronger. The floor arrangement does float higher, about 360 mm of freeboard, because all its foam is under water, so the real trade is freeboard against roll stiffness, and with no roll stiffness to spare freeboard does not help (independent review, LVH-STM-001).

This is the result behind decision D1 in LVH-DDR-001. It matches the practice in the USCG level flotation method, where flotation that keeps a swamped boat upright sits high in the hull.

6. Space, bailing and the dry canoe

Table 6. Space and bailing.

Tag	Quantity	Value
[V1], [V2]	Inside volume below the gunwale; share taken by the modules	5.01 m3; 8.4 % (was 7.6 %)
[B1]	Water inside the swamped canoe	3.48 m3
[B2]	Water that can stay aboard at 200 mm freeboard, all crew and gear aboard	2.31 m3
[B3], [B4]	Water to bail from the lake-level condition; time for four crew with four bailers (D-O3)	1.17 m3; about 2.9 minutes (estimate; 5.9 with two)
	Time to bail the canoe dry at the same rate	about 17 minutes (estimate)
[B5]	Dry canoe: added draft from the 69 kg kit	about 11 mm

The clear width between the module tops is 923 mm (was 963 mm; the model checks at least 900 mm), and the modules take the upper side of the canoe, where little catch is stowed.

7. Fixings

The worst case for a fixing is a module pushed fully under water (a breaking wave or a canoe on its side), in salt water.

Table 7. Fixing loads.

Tag	Quantity	Value	Factor (R7 needs 2)
[S1]	Largest module (1,400 mm), full lift	0.601 kN	
[S2], [S3]	Per strap (two straps); each leg half of that	0.301 kN	Webbing 66
[S4]	D-ring pair, 4 kN to confirm	0.301 kN	13.3
[S5], [S6]	Per coach screw, sideways (largest share, two screws)	0.301 kN	5.0
[S7], [S8]	All modules fully under water; per frame	4.11 kN; 0.205 kN	

The frames take 0.205 kN each into their existing fastenings to the planks. The condition of those fastenings is checked on every hull before fitting (build plan, first checks).

8. Fitting time, cost and the sizing sheet

- [T1] Fitting on the canoe takes about 6.7 hours for a builder and a helper, with the foam cores and covers made beforehand (estimate): marking 40 minutes, battens 120, screws 60, chocks 60 (with the counterbores of LVH-DEC-001 D-S1), straps and modules 50, eye bolts and grab lines 50, marks and bailers 20.
- [C1], [C2] Parts cost USD 694 for one kit and USD 1,384 for two canoes (one gauge), with four bailers a kit (USD 12 more, D-O3). Value-engineering target: USD 1,000. Estimated cost of the constructable design: USD 1,384 (USD 384 over the target). [C3] Foam is 48 % of a kit. The wider modules still come four strips to a 1,000 mm sheet, so the six foam sheets are unchanged; the only added cost is 1 m2 more tarpaulin (USD 5 a kit).

The sizing sheet uses a simple prismatic estimate: modules 220 x 200 mm, top 30 mm below the gunwale, and enough run per side to carry 1.5 times the design net load plus the timber above the water at 50 mm freeboard.

Table 8. Sizing sheet for other hulls (estimates; measure each hull).

Hull	Module run needed per side	Run free of thwarts
6.5 m paddle canoe, no motor, 3 crew	2.5 m	3.8 m
8.0 m canoe, 15 hp, 4 crew (reference)	3.8 m	4.8 m fitted
10 m canoe, 25 hp, 6 crew	5.1 m	6.4 m

9. Results against the requirements

R1 to R13 below are PR-001 to PR-013 of LVH-PRS-001, with the same targets.

Table 9. Results against LVH-PRS-001 (old R numbers).

ID	Target	Result	Status
R1	Afloat with design load	2.45 times the design load carried at 50 mm freeboard	Met on paper
R2	At least 50 mm freeboard	92 mm lowest	Met on paper
R3	Heel and trim within 10 deg	0 deg heel, 0.39 deg trim	Met on paper
R4	Bail in 15 min or less	About 3 minutes from the lake-level condition, about 11 minutes from brim-full, four crew with four bailers	Met on paper (estimate)
R5	10 % or less of inside volume	8.4 %	Met
R6	Under 5 % uptake	Under 2 % by volume on a cited data sheet (about 2 % of lift)	To confirm by test
R7	Fixings 2 times their share	Lowest factor 5.0 (coach screw)	Met on paper
R8	Fitted in a day	About 6.7 hours	Met on paper (estimate)
R9	12 months without loss	Not shown on paper	Needs field trial (TRL 4 and later)
R10	Value-engineering target USD 1,000 for two canoes	USD 1,384	Over the value-engineering target by USD 384
R11	Reserve 1.5	2.45	Met on paper
R12	Afloat with largest module lost	25 mm freeboard (was 9 mm)	Met on paper
R13	Heel within 10 deg, crew on one side	3.9 deg; low side 44 mm (was 4.0 deg, 31 mm)	Met on paper

10. Results added by the STEM check (LVH-STM-001)

These come from `docs/04-calcs/stem_check.py`, which reuses this note's method and model; none changes a result above.

Table 10. Added results.

STEM row	Case	Result	Requirement
S2	Straight after swamping, hull brim-full and the water inside trapped above the lake level	About 24 mm freeboard (reserve 277 kg at zero freeboard over 11.4 m2 of outer waterplane); rises to the 92 mm of [F2] as the water evens out or is bailed	PR-002 (R2), noted
S7	Design net load 20 % higher and foam lift 5 % lower together	76 mm lowest freeboard	PR-002 (R2), met
S18	Partly bailed to 200 mm freeboard, all crew aboard, 2.3 m3 of water still inside	Metacentric height 0.14 m after the free-surface loss of the water inside; crews bail seated	PR-004 (R4), noted
S19	Loaded dry canoe (timber at 750 kg/m3, crew, motor, fuel, gear, catch, kit)	Draft 175 mm; the eye bolt holes at 625 mm are 450 mm above it	PR-017, met
S13	End chock screws after LVH-DEC-001 D-S1	31 mm into the batten (6 d is 30 mm)	PR-007 (R7), met
S31	Wet hull timber denser than water, 1,100 and 1,200 kg/m3 (A2 is neutral, not conservative)	75 and 57 mm lowest freeboard	PR-002 (R2), met
S32	Brim-full, lowest point with the trim of [F3]	About 3 mm (24 mm level)	PR-002 (R2), met on paper as restated for the free-flooding condition, crew bail at once (LVH-DEC-001 D-O2, 2026-10-25)
S33	Roll stiffness of a sealed hull with the water trapped	About 70 kg m per radian brim-full (near neutral), about 1,200 once down to the lake level	PR-003 (R3), to confirm by test
S34	Bailing from brim-full: 2.1 m3 to 200 mm freeboard at a sustained 50 L/min each, four bailers	About 11 min (22 min with two)	PR-004 (R4), met on paper (LVH-DEC-001 D-O3, 2026-10-25)

Two corrections of this note's basis, from the independent review: the lift is taken on the module envelope (0.421 m3) while the foam inside the covers is 0.402 m3; if the sleeves flood, lift is about 4.5 % less, which the 5 % case [F13] (86 mm) bounds. And the "metacentric height" [F5] is the roll stiffness over the net weight carried, not over the whole displacement, so it is not comparable with S18's 0.14 m.

Evidence for the requirements added in LVH-PRS-001: PR-014 (kit inside the hull lines) from the model and LVH-DWG-001; PR-015 (life jacket message) from the safety notes in every document; PR-016 (sizing calculator) from section 8; PR-017 (no new holes below the waterline) from S19 above.